

Calculus II

Integrals of the form $\int \frac{a}{bx^2 + c} dx, c > 0$

Todor Milev

2019

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\int \frac{1}{x^2 + 2} dx$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\int \frac{1}{x^2 + 2} dx$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2 + 1} du = ? + C.$

Example

$$\int \frac{1}{x^2 + 2} dx$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\int \frac{1}{x^2 + 2} dx = \int \frac{1}{2 \left(\frac{1}{2}x^2 + 1 \right)} dx$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\int \frac{1}{x^2 + 2} dx = \int \frac{1}{2 \left(\frac{1}{2} x^2 + 1 \right)} dx$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\int \frac{1}{x^2+2} dx = \int \frac{1}{2(\frac{1}{2}x^2+1)} dx$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned} \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx \\ &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} d\left(\frac{x}{\sqrt{2}}\right) \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned} \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \text{Use } 2 = (\sqrt{2})^2 \right. \\ &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} d\left(\frac{x}{\sqrt{2}}\right) \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned} \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx & \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\ &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} d\left(\frac{x}{\sqrt{2}}\right) \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx & \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right)
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = ? + C.$

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du \\
 &= \frac{1}{\sqrt{2}} ? + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = \arctan u + C$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du \\
 &= \frac{1}{\sqrt{2}} \arctan(u) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = \arctan u + C$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du \\
 &= \frac{1}{\sqrt{2}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{2}} \arctan\left(\frac{x}{\sqrt{2}}\right) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = \arctan u + C$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2+2} dx &= \int \frac{1}{2\left(\frac{1}{2}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2\left(\left(\frac{x}{\sqrt{2}}\right)^2+1\right)} \sqrt{2} d\left(\frac{x}{\sqrt{2}}\right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du \\
 &= \frac{1}{\sqrt{2}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{2}} \arctan\left(\frac{x}{\sqrt{2}}\right) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{u^2+1} du = \arctan u + C$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2 + 2} dx &= \int \frac{1}{2 \left(\frac{1}{2} x^2 + 1 \right)} dx && \left| \begin{array}{l} \text{Use } 2 = (\sqrt{2})^2 \\ \text{Set } \frac{x}{\sqrt{2}} = u \end{array} \right. \\
 &= \int \frac{1}{2 \left(\left(\frac{x}{\sqrt{2}} \right)^2 + 1 \right)} \sqrt{2} d \left(\frac{x}{\sqrt{2}} \right) \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{u^2 + 1} du \\
 &= \frac{1}{\sqrt{2}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{2}} \arctan \left(\frac{x}{\sqrt{2}} \right) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{x^2+1} dx = \arctan x + C$. **Let** $a > 0$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2 + a} dx &= \int \frac{1}{a \left(\frac{1}{a} x^2 + 1 \right)} dx && \left| \begin{array}{l} \text{Use } a = (\sqrt{a})^2 \\ \text{Set } u = \frac{x}{\sqrt{a}} \end{array} \right. \\
 &= \int \frac{1}{a \left(\left(\frac{x}{\sqrt{a}} \right)^2 + 1 \right)} \sqrt{a} du \\
 &= \frac{1}{\sqrt{a}} \int \frac{1}{u^2 + 1} du \\
 &= \frac{1}{\sqrt{a}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{a}} \arctan \left(\frac{x}{\sqrt{a}} \right) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{x^2+1} dx = \arctan x + C$. **Let $a > 0$.**

Example

$$\begin{aligned}
 \int \frac{1}{x^2 + a} dx &= \int \frac{1}{a\left(\frac{1}{a}x^2 + 1\right)} dx && \left| \begin{array}{l} \text{Use } a = (\sqrt{a})^2 \\ \text{Set } u = \frac{x}{\sqrt{a}} \end{array} \right. \\
 &= \int \frac{1}{a\left(\left(\frac{x}{\sqrt{a}}\right)^2 + 1\right)} \sqrt{a} du \\
 &= \frac{1}{\sqrt{a}} \int \frac{1}{u^2 + 1} du \\
 &= \frac{1}{\sqrt{a}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{a}} \arctan\left(\frac{x}{\sqrt{a}}\right) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{x^2+1} dx = \arctan x + C$. **Let** $a > 0$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2+a} dx &= \int \frac{1}{a\left(\frac{1}{a}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } a = (\sqrt{a})^2 \\ \text{Set } u = \frac{x}{\sqrt{a}} \end{array} \right. \\
 &= \int \frac{1}{a\left(\left(\frac{x}{\sqrt{a}}\right)^2+1\right)} \sqrt{a} du \\
 &= \frac{1}{\sqrt{a}} \int \frac{1}{u^2+1} du \\
 &= \frac{1}{\sqrt{a}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{a}} \arctan\left(\frac{x}{\sqrt{a}}\right) + C
 \end{aligned}$$

Linear substitutions leading to blocks IIa and IIIa

Building block IIIa: $\int \frac{1}{x^2+1} dx = \arctan x + C$. **Let** $a > 0$.

Example

$$\begin{aligned}
 \int \frac{1}{x^2+a} dx &= \int \frac{1}{a\left(\frac{1}{a}x^2+1\right)} dx && \left| \begin{array}{l} \text{Use } a = (\sqrt{a})^2 \\ \text{Set } u = \frac{x}{\sqrt{a}} \end{array} \right. \\
 &= \int \frac{1}{a\left(\left(\frac{x}{\sqrt{a}}\right)^2+1\right)} \sqrt{a} du \\
 &= \frac{1}{\sqrt{a}} \int \frac{1}{u^2+1} du \\
 &= \frac{1}{\sqrt{a}} \arctan(u) + C \\
 &= \frac{1}{\sqrt{a}} \arctan\left(\frac{x}{\sqrt{a}}\right) + C
 \end{aligned}$$